

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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Revision: R46, 8 October 2026. **Author:** Qian QI.

We thank the referee for identifying the difference between an intended revision and an actually reviewable source tree. The newest report by repository commit time is the supplemental review of R43 at 9a1ce0502a2cb8f57c697d1df9ba796da0403474, committed at 12:27:38 UTC on 7 October 2026. It reviews the incomplete R43 snapshot 3c9ad7a969bddf3c79ec5dc07e4b74704fed8615, not the completed R45 manuscript. The R44 substantive review at 5d4eca82b5477c4f0305f0f90bf9adc1635ec5de was committed earlier that day, at 10:42:23 UTC. R46 responds to both without treating the numbering of revisions as the chronology of reports.

Our manuscript baseline is the complete R45 at 336ea58decb4fae486efa41bb86e69a28879d9d6. We do not reconstruct an imaginary completed R43, overwrite its historical branch, or revert the manuscript to R41. The title, author, controlled-economy subject, original theory, applications and all adverse experimental evidence remain. The new review-ready branch contains the materialized article, supplement, response, complete new records, generated tables, executable source, tests and compilation audits together.

Beyond the publication repair, the revision adds an action-witness construction. The ReLU minimum circuit carries the original feasible action attaining each label. Its selected-policy residual is bounded directly by the query error. Combining that inequality with the optimal Bellman residual upper bound halves the nonterminal sufficient allowance of the earlier separated construction. The nearest-node neural ablation and conventional spline both receive a strengthened one-sided certificate, so the comparison does not rely on a weak baseline.

The protocol was committed before the new complete execution. All 36 services and 216 rungs were executed, including failures. Every method reaches the targets

1 1/4 and 1/8 in 12/12 services and 1/16 in 6/12. The witness certificate is tighter 1
 2 than the same-critic nearest-node ablation in all 24 distinct economic-cell/resolution 2
 3 comparisons and tighter than the strengthened spline in 23 of 24. The declared 3
 4 first-crossing resolutions are nevertheless identical. Spline is faster in all 30 com- 4
 5 mon successful target/repetition clock comparisons. We distinguish this positive 5
 6 certificate improvement from a policy-cost or speed advantage that these records 6
 7 do not establish. 7

8 1.1. I. Latest R43 supplemental report: blocking concerns 9

10 1.1.1. B1. Incomplete source capsule 11

12 We accept the finding about the pinned R43 snapshot. Its three of five chunks 12
 13 are not presented as a complete submission. R46 uses directly readable UTF-8 13
 14 sources throughout. The final manuscript build does not decode a source capsule 14
 15 or download an expiring workflow artifact. The staging branch is not designated as 15
 16 the referee object. Materialized sources and successful validation precede creation 16
 17 of the new review-ready branch. 17

18 1.1.2. B2. Authoritative entry points remain on R41 19

20 The new branch's root README.md, ECTA.tex and supp.tex select revisions/20
 21 26-10-07-r46/. Its response and numerical summaries are linked from that directory. 21
 22 The earlier root entry point is retained in the preservation audit. The root change is 22
 23 made only when the complete new article and supplement exist. 23
 24

25 1.1.3. B3. Missing article, supplement, response, tables and PDFs 26

27 R46 contains all five classes of objects. The complete R45 article is the baseline; its 27
 28 active labels remain in the revised article and supplement. The witness theorem, 28
 29 policy sandwich, execution allowance and scalar compiler have complete proofs. 29
 30 The existing policy-cost results are retained, not replaced by a protocol. The release 30
 31 audit records the actual compilation of each PDF and rejects undefined references, 31
 32 duplicate labels, missing characters and overfull boxes. 32

1.1.4. *B4. No executed evidence or publication gate*

The new executed study is fixed at source commit f91835f150fdf981f5bab43416121583444a36c7, workflow 37623758585, and evidence commit 257a7b207adf72f07d62d025c93d71a6e1ecb2d2. Its artifact is 11483102637, SHA-256 8cb0fa519bb5e9c15760541d29ca3af5312e73ca728a39ea81a5fc521ba7235a. These identifiers corroborate, rather than replace, the materialized records. code/audit.py reconstructs every checkpoint, formula and first crossing. The R44 diagnostics and direct-comparison records remain in their complete original directory and are covered by inherited regression checks.

1.1.5. *B5. Non-atomic and mutable submission*

The witness-source branch is explicitly a development branch. A new review-ready branch is created only after all materialization, tests, source verification, formula reconstruction and compilation have completed. The publication command refuses to replace an existing review-ready branch. The final commit, rather than an intermediate branch head or an upload message, is the submitted object. Earlier R43, R44 and R45 branches are not updated by this publication.

1.1.6. *B6. Post-review diagnostics cannot change R42 attainment*

The original R42 attainment arrays remain unchanged. The seven R44 residual refinements remain selected post-review diagnostics of frozen actors. The R45 deterministic construction remains its own finite catalogue. R46 is a fourth, prospectively specified service catalogue with a different witness actor and strengthened comparator certificates. Its 12/12 attainment at 1/8 does not retrospectively turn the original R42 or R45 capped failures into successes. Nor is the witness actor called an unchanged-policy recertification.

1.1.7. *B7. Paired simulation needs a theorem and implementation audit*

The active supplement retains the full paired score, deterministic support, interval-valued endpoint argument, outward moments and familywise concentration proof. Under expectation, telescoping identifies actual fixed-policy cost. The lower and upper endpoint sequences enclose the original continuous innovation

law; the 24 fixed comparisons use two one-sided tails each and a total error budget 0.01. Actor-cell ambiguities, terminal terms and numerical error remain explicit. The new one-sided policy sandwich is an order theorem and is not used to extend the expectation-specific direct cost identity to nonlinear certainty equivalents.

1.1.8. B8. *Twenty-four state comparisons do not imply method reliability*

We retain the conditional fixed-policy interpretation. The new construction has a deterministic termination theorem under stated effective-computability conditions; its finite-cap outcome is a separate object. Three timing repetitions produce identical policy/certificate checkpoints and do not estimate initialization randomness. There is no inference from four initial states or from repeated deterministic runs to a distribution of economic tasks.

1.1.9. B9. *Finer verification is not nonlinear scaling*

The witness theorem is finite-dimensional and reduces the sufficient nonterminal resource allowance. The scalar witness compiler additionally supplies a linear-size exact representation with an action index on each cell. We display the remaining state/action covering factors and the rational-arithmetic cost. The executed study is scalar, varies horizon and price, and is not described as a genuinely high-dimensional nonlinear benchmark. The earlier two-state and structured quadratic experiments retain their distinct meanings.

1.1.10. B10. *Substantive R42 findings remain*

All original comparator and precision findings remain in the main text. R46 adds a measurable positive result: the witness policy has a tighter certified bound in every same-critic ablation comparison and in 23 of 24 strengthened-spline comparisons. It also retains the unfavorable comparator exception, all tight-target failures and every clock favoring spline. A narrower certified regret interval does not establish lower actual expected cost. Existing unresolved direct-cost intervals are not relabeled as superiority or equivalence.

1.2. II. Latest R43 supplemental report: major comments

1.2.1. M1. One immutable review-ready commit

The new branch is published from one complete materialized tree after the gates, with a distinct name and final commit. It is not the former R43 staging branch. All work-in-progress commits remain on the source branch, and the evidence branch is separately identified.

1.2.2. M2. Materialize the source tree

The article, supplement, response, sections, tables and Python programs are ordinary source files. An ordinary repository checkout contains the fixed dependencies in R44/R45 alongside R46. There is no capsule restore step in the review build. A manifest hashes actual files, not only a compressed container. The incomplete historical R43 capsule remains historical evidence and is not advertised as the reproduction route.

1.2.3. M3. Update authoritative entry points

Root entry points select R46 and the new README links its three PDFs, raw records, numerical summary and release audit. The manuscript separates original capped-service outcomes, post-review diagnostic outcomes and the two constructive catalogues. The review chronology and exact underlying commits are recorded in this response and the protocol.

1.2.4. M4. Commit the seven diagnostic records separately

The complete seven R44 recertification records, their original allowance accounts, additional resolutions, actor/network identities, incremental work and clocks remain in revisions/2026-10-07-r44/results/. Their separate diagnostic tables remain active in the article and supplement. The inherited tests reconstruct their bounds and verify the original actor and network hashes. We do not rerun these old clocks or rename them R46 prospective observations.

1.2.5. *M5. Commit raw direct-comparison records*

All 24 original paired records retain their identities, states, seeds, sample counts, endpoint statistics, support intervals, numerical allowances and confidence endpoints in the R44 result directory. Their full generated table remains in the active supplement. The new revision does not replace raw comparison evidence with an assertion based on regret bounds. No new direct policy-cost simulation is claimed in the witness catalogue.

1.2.6. *M6. Prove the direct difference in the supplement*

The active supplement contains the retained complete expectation-specific proof. The simultaneous family is fixed at 24 policy/state comparisons and 48 one-sided tails. The result is conditional on the frozen policies and the stipulated independent simulation model; deterministic pseudorandom reproducibility is not a proof of that model's independence. The new policy sandwich has its own complete recursive order proof and comparison class.

1.2.7. *M7. Keep clocks nonadditive across environments*

Historical clocks are not summed with new clocks. The new services have their own complete prefix and process accounts on one runner. Original diagnostic work and paired-simulation work retain separate identities. The pooled clock ratio in the machine-readable summary is explicitly descriptive: target prefixes overlap, so it is neither an independent-sample statistic nor a total-study runtime. Construction counts and storage are not mislabeled as FLOPs.

1.2.8. *M8. Preserve every adverse result*

All historical evidence is retained. R46 also preserves the witness/spline exception at horizon 4, price 1, $N=16$; the six failures per method at $1/16$; and spline's faster clocks in all 30 common successful comparisons. Target decisions use exact rational bounds, not displayed rounded decimals. No equivalence margin is chosen after seeing a cost interval.

1.2.9. M9. Address method-level questions explicitly

The witness theorem supplies a constructive policy guarantee with a smaller sufficient resource account and a precise numerical-selection allowance. The full matched study supplies finite work-to-target and certificate frontiers for the specified generators. Generic nonlinear optimizer reliability, genuinely multidimensional competitiveness, mixed-precision work savings and calibrated economic discovery remain separate substantive questions, not conclusions manufactured from the new scalar evidence. The original NBO research program is advanced by a new construction rather than abandoned because those comparisons are not all established.

1.2.10. M10. Focus without replacing the paper

We retain *Neural Bellman Operators* as the central paper. The organizing object is the policy-specific continuation together with its feasible improvement, certificate and resource account. The new witness construction is a result about that same object. Controlled diffusions, recursive preferences, temporal selves and games remain with their original hypotheses and full preserved developments, not as purported new empirical successes of the scalar experiment. The title and economic subject are unchanged; no theory or application is deleted to manufacture a different paper.

1.3. III. Continuing response to the R44 substantive report

B1 and M1 (distinctive construction). The new actor is tied to the attaining cone, yielding a selected-policy residual controlled only by query error. The theorem does more than accept a previously fitted candidate. Classical envelopes and elementary ReLU gates are credited; an equivalent non-neural implementation remains possible. The increment is the witness-preserving own-future policy/resource account and its isolated ablation.

B2, B8, M2 and M3 (prospective service and reliability). The pre-execution protocol fixes all generators, economic cells, ladder, targets and repetitions. Every service is executed. Deterministic termination is the theoretical reliability object; finite

capped attainment is reported separately. No generic Adam or L-BFGS guarantee is inferred.

B3, B6, M4 and M7 (common frontier and complete work). Both nearest-node baselines receive the stronger certificate. All 72 distinct method/economy/resolution rows, their three repetitions, witness storage, primitive queries and prefix clocks are recorded. Equal first-crossing resolutions and spline's faster clocks remain visible. Additional certificate sharpness is not renamed a work-to-target advantage.

B4 and M5 (direct policy values). The old 24 intervals remain unresolved. We neither invent a new superiority result nor choose an ex post equivalence margin. A future direct comparison must be designed for that estimand; it is not silently included in the current construction clock at zero cost.

B5 (strong conventional evidence). The new witness bounds improve on the strengthened spline in 23 of 24 unique comparisons, which is a genuine improvement of this certificate frontier. The one exception, unchanged crossing resolutions, and slower neural clocks preclude a blanket dominance claim. All previous adverse comparisons are preserved.

B7 and M6 (dimension). The theorem and actor are finite-dimensional; the executed exact compiler is scalar. Neither is used to claim removal of tensor covering costs. No unexecuted sparse-grid, fitted-value or policy-iteration benchmark is reported as a completed comparison.

B9 (precision). The existing precision acceptance theory and adverse measured comparison remain unchanged. The witness implementation uses exact rational compilation and outward arithmetic, not a newly successful mixed-precision speed experiment.

B10 and M10 (economic contribution). The policy tolerance and execution allowances remain in discounted economic-cost units and support a well-defined guarantee in the original nonlinear investment laboratory. We do not describe this laboratory as calibrated or assert that its conclusions are inaccessible to conventional computation. The broader economic applications remain preserved and conditionally stated.

M8 (recursive versus expected costs). The one-sided sandwich uses order and cash invariance; the paired direct-cost score uses conditional expectation. Both statements and their assumptions are separately proved. The witness result does not broaden the statistical score's scope by analogy.

M9 (self-contained submission). The active article and supplement include the new theorem chain and all tables needed to interpret the new execution. Their sources compile from one repository snapshot. Older, extensive applications remain physically preserved and available with their original conditions. The latest review is included as source material, not conflated with a review of R45.

1.4. IV. Validation and reading map

The publication build reruns 14 new exact regressions, 13 inherited constructive regressions and 14 inherited R44 regressions, without modifying historical result files. It reconstructs 216 new checkpoints, all exact bounds and first crossings, identical repetition objects, and identical critics in the two neural actor variants. It compiles all three documents with the retained Econometric Society class and rejects unresolved references, duplicate labels, missing characters and overfull boxes. These checks are reproducibility evidence, not mathematical peer approval or an editorial decision.

Read the new main section *Neural continuation with an action witness*, its complete proof section in the supplement, the full frontier and timing tables, audit/PUBLICATION_SUMMARY.json, audit/RESULT_AUDIT.json, and audit/RELEASE_AUDIT.json. The final branch and commit bind those objects. No earlier review or revision is overwritten.

1.5. V. State-dependent feasibility and final integration

The added main section, *State-dependent constraints and feasible witnesses*, and its proof supplement address the constructive gap between the original feasible correspondence and the common-action backend. Under a Hausdorff-Lipschitz modulus, feasible-pair primitive moduli, effective node-specific action nets and feasible Borel repairs, the theorem constructs the same min-plus ReLU continuation

and a repaired feasible policy. The policy loss charges action-net error, query error, state coverage and repair displacement. It reduces to the common-action witness theorem when constraint variation and repair error vanish. A capacity-constrained two-state example has an exact clipping repair. Eight independent exact rational tests check its admissibility and bound formulas; the analytic proof establishes the all-state result. This is additional mathematical scope, not evidence of high-dimensional comparative speed or a calibrated application.

The inherited R46 publication attempt 37626921466 stopped because its label-preservation traversal preceded generation of the tables it traversed. We generate and audit the actual tables first; no missing input, test, or label check is waived. A separate source branch holds the integration, and only a successful complete build is promoted to the new review-ready branch dated 8 October 2026. The final build runs 49 tests, preserves the old 36-service/216-rung catalogue and all earlier result paths, and includes the complete latest R43 report. The former failed run remains a failed run.